Week 3 cone pusher preparation

Robot dance studio + removable cone pusher / 12 children / four teams

Print

[ ] 5 team sheets + 4 build cards + 1 attendance sheet = 10 required pages. Reuse role cards and hub guides. This checklist and a lead note copy are optional.

Pack

[ ] Reusable kit: 2 assembled robots, 4 laptops/chargers, tested cables, 14 pencils/erasers, 4 rulers, 4 team trays and 2 timers.

[ ] 6 matching lightweight cones with flat bases (suggested height 4-8 cm): 4 team props and 2 robot-lane props. Check that they slide on the actual surface.

[ ] 4 equal pusher build packs, 4 mounting proxies and 2 mentor pushers. Beams/pins or pre-cut card, two mounting supports and braces per team, craft tape. Keep both bases assembled.

[ ] Floor-safe tape for two clear lanes, starts and reachable target zones; 10 cm hand-test paths and zones wider than the cone at build tables.

Rehearse on BOTH kits

[ ] Read the full W03_dance_Checklist.md. Test original dance with pusher removed. A left/B right; auxiliary motor unplugged; no sensors required. Check Stop and save workflow.

[ ] Fit on stopped base; clear wheels, floor, cables and buttons. Test quick swaps. Cone loop: three repeats of await drive(60), await pause(300); await signal("D") outside. SPEED = 200; no turns.

[ ] Mark target for actual 180-wheel-degree travel. Rehearse three runs. Success: upright, whole cone base inside zone. Stop for tipping/escape; reset only after motion stops. Record actual results.

Teach and close

[ ] Hand-test three times, change one feature, repeat three times. Keep cone, floor, start and travel fixed. Rotate builder, designer and tester.

[ ] Use 38-50 minute retests for fit and cone loop. Preserve each driver turn and minute-50 reflection. Use mentor pusher if team fit fails; share hand-test evidence.

[ ] Save team projects, label four builds, count parts and record shortages/results in SESSION_PREP.md. Keep completed child records private.